AUTONOMOUS STRUCTURAL SCREENING

# TwinVerse Inspect AI

## Inspect the unreachable.

Drone, CCTV and handheld imagery in — located defects, ranked severity, and a shareable report out.


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[ Event / Track ] · [ Date ] · Bano Qabil · Alkhidmat Foundation Pakistan

THE PROBLEM

# Inspection today is manual, slow, and genuinely dangerous.

### Dangerous

Engineers rappel down dams and climb bridge soffits to look at concrete with their own eyes.

### Slow

A single span can take days to survey. Backlogs mean structures go years between inspections.

### Inconsistent

Two inspectors, two clipboards, two different answers about the same crack.

### Reactive

Defects are found after they matter, not before. Maintenance is repair, not prevention.

WHAT WE BUILT

# A first pass that never has to climb anything.

**01**

### Capture

Drone, CCTV, robot or a phone. Images or video.

**02**

### Detect

A fine-tuned YOLOv11 model locates cracks and draws a box around each one.

**03**

### Score

Every detection gets a severity score from a formula shown on screen.

**04**

### Report

Annotated images, a 3D view, and a PDF anyone can forward.

*The engineer still decides. The system removes the part of the job that is climbing, photographing, and sorting.*

STATUS

# Not a mockup. A running system.

**104**
automated tests

**23**
endpoints

**6.1s**
to analyse 8 images

**7ms**
inference per image

- Upload API with S3-compatible object storage
- Severity engine with a formula served live to the UI
- PDF reports with a limitations page
- Fine-tuned crack detector, trained on 1,317 images
- Next.js dashboard with 3D digital twin viewer
- JWT auth, three roles, Docker Compose, CI

EXPLAINABILITY

# Every number can be recomputed by hand.

```
severity = area × confidence × class weight
```

A real detection from the demo data:

```
0.0288 × 0.558 × 1.0 = 0.01606
```

The formula, the weights and the band thresholds are served by the API and rendered live — so what the dashboard shows can never drift from what the server computed.

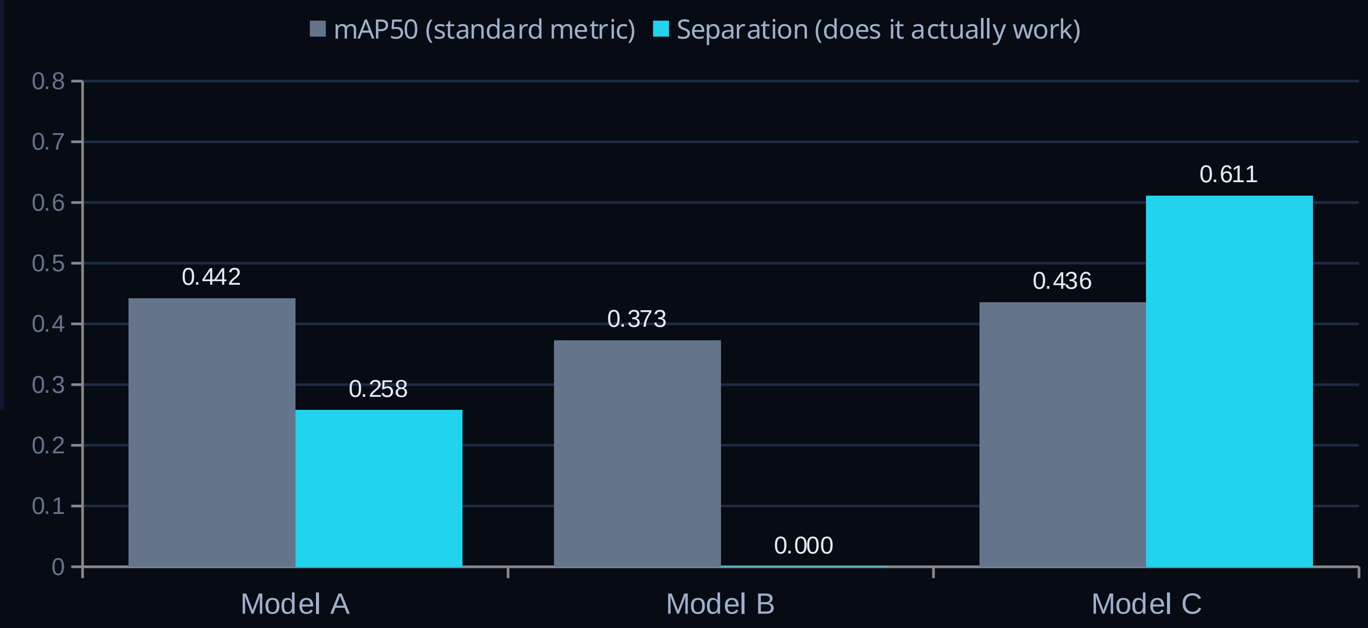
SEVERITY BANDS

| | |
|---|---|
| Critical | > 0.014 |
| High | 0.011 – 0.014 |
| Medium | 0.009 – 0.011 |
| Low | < 0.009 |

*Calibrated against 308 real detections, not assumed.*

THE MEASUREMENT THAT MATTERED

# The standard metric said all three models were the same. They were not.




### What we learned

mAP is computed only over defects that are labelled. A validation set with no clean images cannot catch a model that fires on everything.

So we tested each model against 94 defect-free photographs.

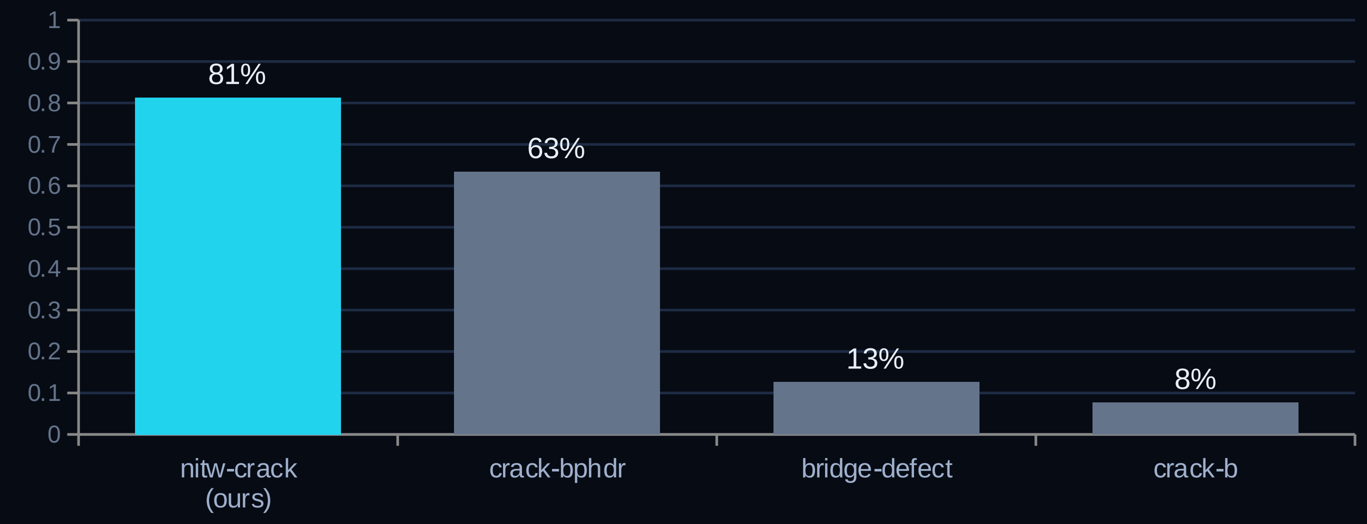
Model A looked fine and flagged half of all clean concrete. Model C flags one in five.

Then we asked the harder question: does it hold up on imagery from somewhere else?

THE NUMBER MOST TEAMS NEVER CHECK

# Then we tested it on three datasets it had never seen.

Every accuracy figure above came from our own dataset's held-out split — different photographs, but the same cameras and the same walls. So we ran it against three public datasets from unrelated sources, none of which contributed anything to training.




### What this means

Four in five cracks on imagery like our training set. On three we had never seen, between 63% and 8%.

We could have shown you only the 63%. We are showing you all four, because the honest claim is that this is tuned to one kind of concrete photography.

*We would rather tell you this than have you find it.*

VISUALISATION

# From a photograph to a structure you can orbit.

### On the image

- Bounding boxes drawn over each detection
- Colour and size both carry severity
- Hover a finding to see its arithmetic
- Boxes stored normalised, so they align at any zoom

### Digital Twin v1

- A 3D structure with glowing severity markers
- Click any marker for its source image and score
- Built in code — no scanning rig required
- Illustrative placement, and we say so on screen

*[ Replace this line with a screenshot of the dashboard and the 3D viewer ]*

WHAT IT DOES NOT DO

# We measured the weaknesses too.

Every one of these is stated in the product itself and on the last page of every report.

### Cracks only

One defect class. It has never seen corrosion or spalling and will not report them.

### 1 in 5 clean surfaces flagged

Measured against 94 defect-free photographs. It errs toward flagging.

### 8-63% on unfamiliar imagery

Versus 81% on imagery like its training data. Three unseen datasets, all published.

### Severity is relative

It ranks which crack to look at first. It does not measure width in millimetres.

### The 3D view is illustrative

Marker positions come from capture order, not surveyed coordinates.

### Video counts are inflated

Frames are analysed independently, so one crack can be counted many times.

### Not an engineering assessment

It is a screening pass. A qualified engineer still signs off.

*Naming a weakness costs far less than being caught by it.*

UNDER THE HOOD

# Boring technology, deliberately.

| | |
|---|---|
| **Frontend** | Next.js 16 · React 19 · Three.js · Tailwind |
| **API** | FastAPI · Python 3.12 · SQLAlchemy · Alembic |
| **Model** | YOLOv11n fine-tuned · PyTorch 2.13 · CUDA 13 |
| **Data** | PostgreSQL 16 · MinIO (S3-compatible) |
| **Ship** | Docker Compose · GitHub Actions CI · JWT + RBAC |

### Decisions on record

20 architectural decisions are written down with their reasoning — including the ones we got wrong, and one we had to publicly reverse after re-checking it.

A dataset that looked ideal and turned out unusable. A merge that made the model worse. A metric that hid a failure for an hour.

All of it is in the repository.

THE TEAM

# Three people, one build.

PHOTO

### Ayaan Aatif

Team Lead

Architecture, ML pipeline, demo delivery

PHOTO

### Muhammad Muneed

Team Member

[ Add focus area ]

PHOTO

### Inshrah Mehmood

Team Member

[ Add focus area ]

*[ Add contact email · GitHub · LinkedIn ]*

WHAT COMES NEXT

# The honest roadmap.

### SHIPPED

- Crack detection, measured on unseen data
- Severity scoring and PDF reports
- Dashboard, 3D viewer, auth, Docker

### NEXT

- Varied training data to close the 63/81 gap
- More defect classes — corrosion, spalling
- Cross-frame tracking so video counts are real

### LATER

- Photogrammetric twin from the imagery itself
- Predictive maintenance once trend data exists
- Live drone and robot integration

*Predictive maintenance needs longitudinal data that does not exist yet. We are not going to pretend otherwise.*

# This does not replace an engineer.

It does the first pass, so the engineer spends their time on judgement instead of data collection — and it tells you exactly how much to trust it.

**TwinVerse Inspect AI**

Ayaan Aatif · Muhammad Muneed · Inshrah Mehmood

*[ Add repository link · contact email ]*